TrES-2b

R-1111

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_140516` · created 2026-07-30T20:49:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456793.89917 +/- 0.00029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.014
Transit depth (Rp/Rs)^2	1.91 +/- 0.38 [%]
Orbital Inclination (inc)	84.22 +/- 1.38
Airmass coefficient 1 (a1)	0.9249 +/- 0.0091
Airmass coefficient 2 (a2)	0.069 +/- 0.023
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#1 - [159, 412]
Optimal Aperture	3.55
Optimal Annulus	10.06
Transit Duration (day)	0.078 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.014 vs 0.1254 (deviation 0.72σ)
Duration fit vs published	0.078 d vs 0.0746 d (deviation 0.12σ)
Mid-time O–C	-0.7 min
Depth SNR	7.9
Residual scatter	0.99 %

Light curve

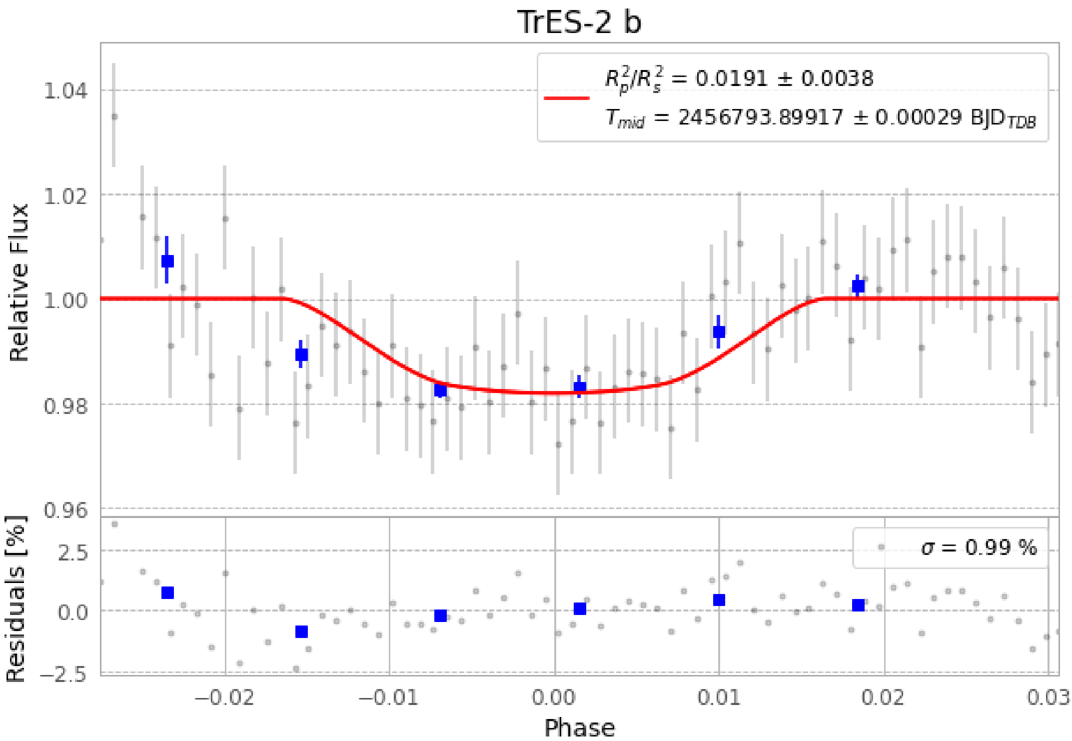

Figure 1 — FinalLightCurve_TrES-2 b_2014-05-16.png (SHA-256 802cb5952f7067db…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 281], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4bc615dd9f84e75a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

71 FITS frames (47 MB), `TRES-2140516075112.FITS` → `TRES-2140516112112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-140516.log (f2a9204d43bb…), FinalLightCurve_TrES-2 b_2014-05-16.png (802cb5952f70…), FinalLightCurve_TrES-2 b_2014-05-16.csv (f015a364e969…)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-30 20:49Z.